Particle Physics 2: The Standard Model
Lecture 9: Fermion Masses and Yukawa Couplings

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Chapter 1

Fermion Masses: Chirality, Yukawa Couplings, and Majorana Neutrinos

Overview. We first rebuild the gauge-boson Higgs mechanism in a toy $U(1)$ theory, then ask the parallel question for fermions. A Dirac mass mixes left and right chiral fields, but the weak interaction assigns them different charges and therefore forbids a bare mass. A Yukawa coupling to the Higgs field respects the symmetry; after the Higgs field acquires a vacuum value, the same interaction becomes a fermion mass and a measurable Higgs–fermion coupling. The final part follows this logic into neutrino mass, possible heavy unprotected particles, cosmic-ray kinematics, relativistic inertia, and gravity.

The purpose of the opening recap is not merely to repeat the previous lecture. We need one idea from it in a form that can be carried over to fermions: a mass is a nonzero energy for an excitation at zero momentum. For a field, zero momentum means infinite wavelength, or a displacement that is the same everywhere. A derivative-only theory cannot detect a uniform displacement. A term without derivatives can.

1.1 The Gauge-Boson Story in One Pass

1.1.1 Derivative energy and a massless vector

The model gauge field is A_μ, with field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{1.1}$$

The board writes the gauge kinetic term schematically as F^2. With a standard normalization it is

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \tag{1.2}$$

Every appearance of A_μ in (1.2) is differentiated. Consequently,

$$A_\mu(x) \longmapsto A_\mu(x) + c_\mu, \qquad \partial_\nu c_\mu = 0, \tag{1.3}$$

does not change $F_{\mu\nu}$. A homogeneous vector potential therefore costs no field-strength energy. In the language of the lecture, the zero-momentum mode has no gap and the gauge boson is massless.

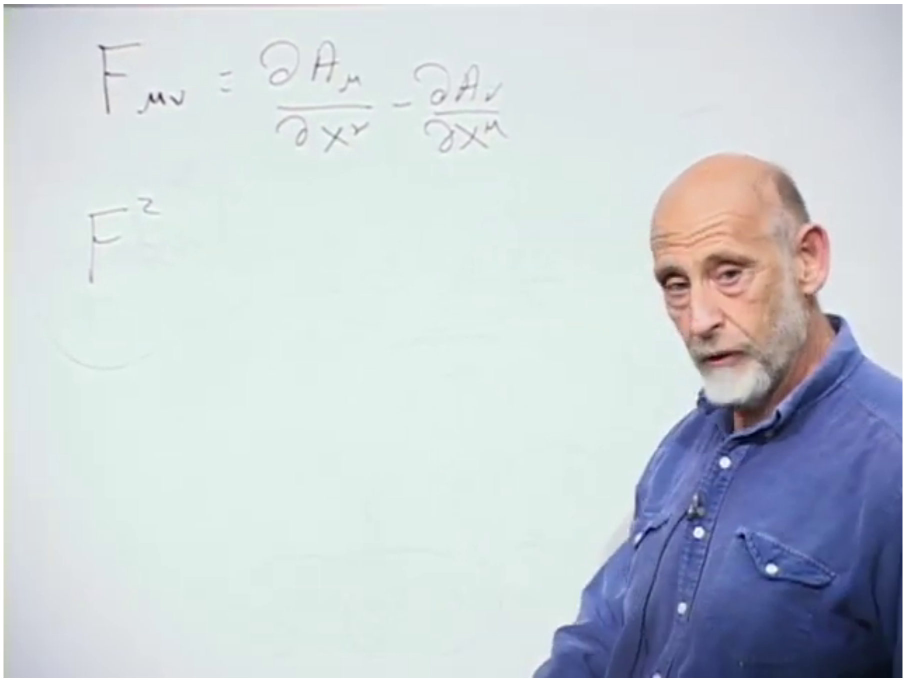

Figure 1.1: The field-strength definition and the board's F^2 shorthand for the derivative energy of the gauge field.

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This $U(1)$ field is a deliberately simple stand-in. The physical target is the $W^\pm$ and Z, not the electromagnetic photon. The unbroken electromagnetic symmetry leaves the photon massless; the broken electroweak symmetry gives masses to the weak gauge bosons.

1.1.2 Freezing the Higgs radius

Introduce a charged complex scalar and write it in polar form,

$$\phi(x) = \rho(x)e^{i\alpha(x)}. \tag{1.4}$$

Its Mexican-hat potential is minimized on a circle of nonzero radius. Calling that radius f, the radial direction is comparatively stiff. For the first pass we freeze it and retain only the phase:

$$\phi(x) \simeq f e^{i\alpha(x)}. \tag{1.5}$$

Choose the convention

$$D_\mu = \partial_\mu + igA_\mu, \qquad \phi \longmapsto e^{i\theta}\phi, \qquad A_\mu \longmapsto A_\mu - \frac{1}{g}\partial_\mu\theta. \tag{1.6}$$

The lecture suppresses g; keeping it visible makes the dimensions and the eventual mass transparent. Acting on the frozen field gives

$$\begin{aligned} D_\mu\phi &= (\partial_\mu + igA_\mu)fe^{i\alpha} \\ &= i(\partial_\mu\alpha + gA_\mu)fe^{i\alpha}. \end{aligned} \tag{1.7}$$

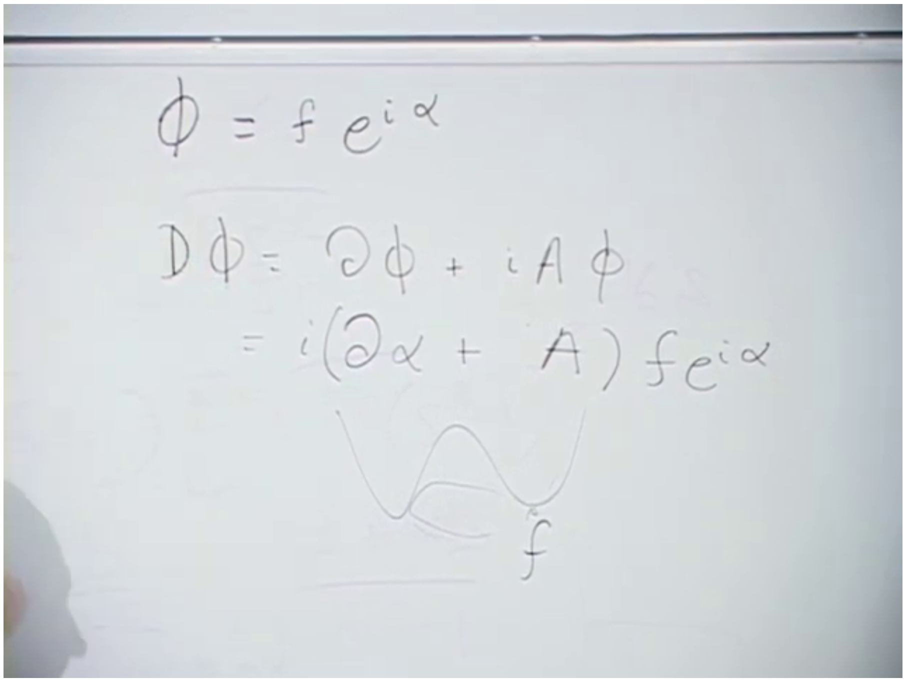

Figure 1.2: The frozen Higgs radius, its covariant derivative, and the small broken-potential sketch used in the recap.

Lecture frame: 00:06:42.

Multiplication by the complex conjugate removes the phase and also turns i times $-i$ into $+1$:

$$|D_\mu\phi|^2 = f^2(\partial_\mu\alpha + gA_\mu)(\partial^\mu\alpha + gA^\mu). \tag{1.8}$$

Question from the audience. Should the square of the covariant derivative carry a minus sign because the derivative produced a factor of i? [a]

Response. No. The kinetic term is the product with the complex conjugate. One factor contributes i, the other $-i$, and $i(-i) = 1$. The sign convention for D_μ may be reversed consistently, but the modulus square is positive in the corresponding energy expression.

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[a] Lecture timestamp: 00:07:39.

The combination in (1.8) is itself a shifted gauge field. Choose $\theta = -\alpha$, so that the scalar phase disappears, and define

$$A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha. \tag{1.9}$$

Then

$$|D_\mu\phi|^2 = g^2 f^2 A'_\mu A'^\mu. \tag{1.10}$$

The new term contains no derivative of A'_μ. A homogeneous field now stores energy, so the zero-momentum vector mode is gapped.

With canonical scalar normalization one writes $\phi = (f+h)e^{i\pi/f}/\sqrt{2}$. Comparing the resulting vector term with $\frac{1}{2}m_A^2 A_\mu A^\mu$ gives

$$m_A = gf \tag{1.11}$$

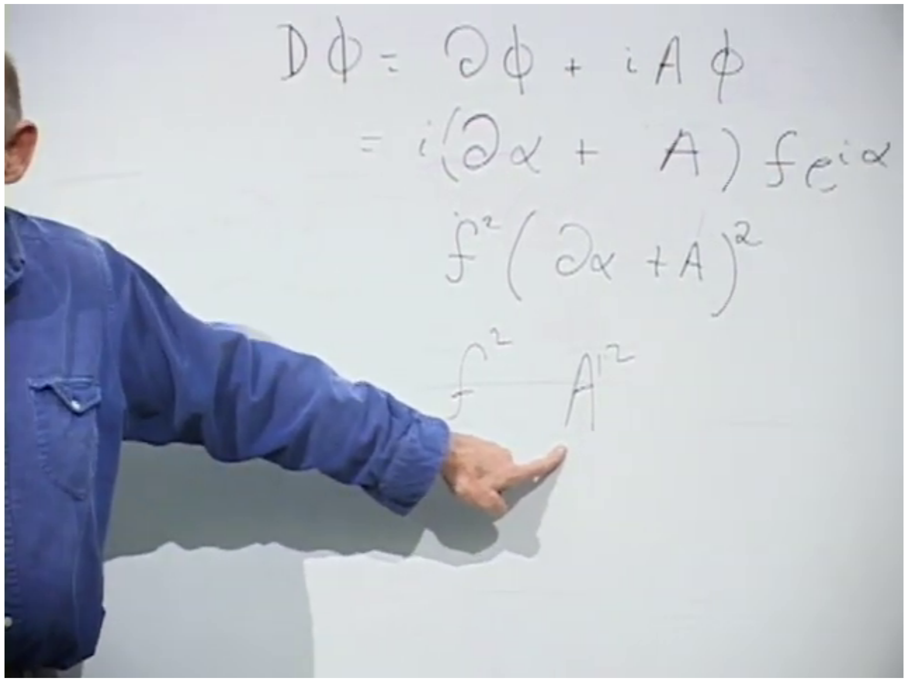

Figure 1.3: The board isolates the $f^2 A^2$ term that signals a gauge-boson mass after the phase has been absorbed.

Lecture frame: 00:10:31.

in this abelian model. The lecture suppresses the normalization factors and summarizes the result by saying that the vector mass is set by f.

The phase mode has not vanished without accounting. Before gauging, it was a massless Goldstone boson. After gauging, it supplies the longitudinal polarization needed by a massive vector particle. The radial fluctuation, which we temporarily froze, remains as the physical Higgs excitation.

1.1.3 Why a term without derivatives means mass

Question from the audience. Why does an A'^2 term count as a mass rather than just another interaction?[a]

Response. Mass is energy at rest. For a field, a particle at rest is the quantum of a zero-momentum, infinite-wavelength mode. The derivative term gives no energy to a static homogeneous displacement, whereas $m_A^2 A'^2$ does. That nonzero zero-momentum energy is the rest-energy gap.

[a] Lecture timestamp: 00:13:10.

The lecture develops this statement through three mechanical analogies.

- In a crystal, slowly varying displacement costs elastic energy because neighboring atoms are shifted differently. A rigid translation shifts every atom together and costs no internal energy. The acoustic phonon is therefore gapless.
- In an ideal rotationally symmetric magnet, a spatially varying spin direction costs exchange energy, but rotating every spin together does not. The associated long-wavelength spin wave is gapless.

- In a plasma, shifting the electron fluid relative to the positive background creates charge separation and a restoring electric field even for a homogeneous relative displacement. The resulting plasmon has a nonzero frequency at zero wave number.

The Higgsed gauge field behaves like the third example: the broken medium resists what would otherwise be a cost-free homogeneous displacement.

Question from the audience. Before the gauge field was introduced, did the phase α not already vibrate?[a]

Response. It did. Its energy was proportional to $f^2(\partial_\mu \alpha)^2$. A phase that varies from place to place has gradient energy and can propagate, but a uniform change of phase costs nothing. That is precisely the massless Goldstone mode.

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[a]Lecture timestamp: 00:18:24.

Question from the audience. The derivative includes a time derivative. Does actually moving the whole lattice or changing the whole field not carry kinetic energy?[a]

Response. Yes. The comparison is between static configurations, or between configurations prepared arbitrarily slowly. A lattice translating with nonzero velocity has kinetic energy. The mass test asks whether the configuration still has energy when it is at rest and spatially uniform.

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[a]Lecture timestamp: 00:22:18.

1.2 The Experimental Clue: Weak Interactions Distinguish Left and Right

The lecture now turns to the electron, muon, and quarks. Their masses also come from the Higgs field, but the mechanism looks different because a fermion mass is not a quadratic vector term. It is a coupling between two chiral fields.

1.2.1 Parity violation in beta decay

Begin with the empirical fact that the weak interaction violates parity. Quantum electrodynamics and quantum chromodynamics treat a process and its mirror image alike. The weak interaction does not. In neutron beta decay,

$$n \longrightarrow p + e^- + \bar{\nu}_e, \tag{1.12}$$

the high-energy electrons selected by the charged-current interaction have left-handed helicity: their spin points opposite to their momentum. The corresponding antineutrinos have the opposite helicity.

For an ultrarelativistic particle, define helicity by

$$h = \frac{\boldsymbol{S} \cdot \boldsymbol{p}}{|\boldsymbol{S}|\,|\boldsymbol{p}|}. \tag{1.13}$$

Spin parallel to momentum has $h = +1$; spin antiparallel has $h = -1$. The weak interaction couples to a left-chiral electron field and a left-chiral neutrino field, not to a right-chiral electron field in the same way.[1]

To expose the obstruction without the full $SU(2)_L \times U(1)_Y$ bookkeeping, imagine a fake $U(1)$ world in which only ψ_L carries the relevant charge:

$$\psi_L \longmapsto e^{i\theta}\psi_L, \qquad \psi_R \longmapsto \psi_R. \tag{1.14}$$

This is not electromagnetism: real left- and right-chiral electrons carry the same electric charge. It is a caricature of the weak interaction, in which the two chiralities sit in different electroweak representations.

Question from the audience. Does the same left–right asymmetry apply to quarks? [a]

Response. Yes, with the appropriate weak multiplets and hypercharges. Left-chiral up- and down-type quarks form weak doublets, whereas the right-chiral fields are weak singlets. Their Higgs couplings therefore face the same structural problem as the electron's.

[a]Lecture timestamp: 00:31:50.

1.3 A Dirac Mass Is Left–Right Mixing

1.3.1 From the four-component equation to two Weyl equations

A Dirac wavefunction has four components. Loosely, two describe spin and two distinguish the positive- and negative-energy solutions, with the latter reinterpreted as antiparticle degrees of freedom. In Hamiltonian form,

$$i\frac{\partial \Psi}{\partial t} = (-i\boldsymbol{\alpha}\cdot\boldsymbol{\nabla} + m\beta)\,\Psi. \tag{1.15}$$

The matrices α_i and β encode the relativistic relation between energy, momentum, and mass.

The chiral basis makes the role of m explicit. With $\sigma^\mu = (1, \boldsymbol{\sigma})$ and $\bar{\sigma}^\mu = (1, -\boldsymbol{\sigma})$,

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R, \tag{1.16}$$
$$i\sigma^\mu \partial_\mu \psi_R = m\psi_L. \tag{1.17}$$

At $m = 0$, the equations are independent: a left-chiral field and a right-chiral field can propagate without talking to each other. At $m \neq 0$, each appears as the source for the other. This is the exact sense in which a Dirac mass mixes left and right.

Susskind describes this as the fermion oscillating between left and right. That language captures the coupling but should not be pushed into a literal classical picture of a point particle flipping back and forth. The precise statement is the coupled field equation (1.16)–(1.17).

Question from the audience. Are the factors of i and the signs in the two board equations consistent?[a]

Response. Not line by line on the board; an i was pulled outside inconsistently during the quick derivation. The reliable content is the coupling pattern. Equations (1.16) and (1.17) give one standard consistent convention.

[a]Lecture timestamp: 00:41:10.

[1]Editorial clarification: The lecture initially uses handedness in the helicity sense because that is visually intuitive at high energy. Gauge charges actually act on chirality, the Lorentz-representation label selected by $P_L = (1-\gamma^5)/2$ and $P_R = (1+\gamma^5)/2$. For a massless particle, chirality and helicity coincide for particles; for a massive particle they do not.

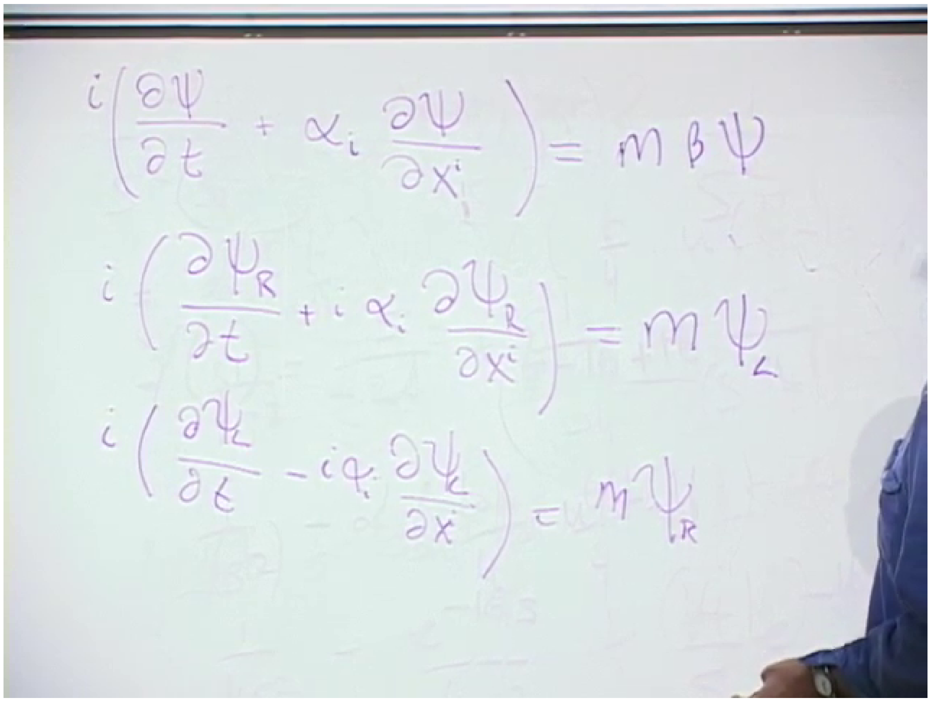

Figure 1.4: The board first writes the Dirac equation, then splits it into coupled left- and right-handed equations.

Lecture frame: 00:40:30.

1.3.2 Why the weak symmetry forbids a bare mass

The corresponding Dirac mass term in the Lagrangian is

$$\mathcal{L}_m = -m\bar{\psi}\psi = -m\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right). \tag{1.18}$$

Under the toy transformation (1.14),

$$\bar{\psi}_L\psi_R \longmapsto e^{-i\theta}\bar{\psi}_L\psi_R, \qquad \bar{\psi}_R\psi_L \longmapsto e^{i\theta}\bar{\psi}_R\psi_L. \tag{1.19}$$

Neither term is invariant. The direct left–right mixing coefficient m is therefore forbidden before symmetry breaking.

Electromagnetism alone would not cause this problem, because ψ_L and ψ_R have the same electric charge. The obstruction arises from their different weak quantum numbers.

1.3.3 Helicity questions that expose the chirality subtlety

The class now tests the intuitive handedness picture from several angles. These questions are not side issues: they reveal exactly where helicity is useful and where chirality must replace it.

Question from the audience. What would the world be like if the weak interaction emitted both electron helicities equally?[a]

Response. The immediate parity asymmetry would disappear from that process. But the real electroweak theory is chiral: only the left-chiral doublets carry the charged weak interaction. The lecture postpones a fuller account of why nature chose that structure.

[a]Lecture timestamp: 00:42:12.

Question from the audience. Suppose an electron's spin is prepared vertically and the electron is then accelerated horizontally. Is it left- or right-handed? [a]

Response. A spin state transverse to the momentum is a superposition of the two helicities. Helicity is sharp only when the spin is parallel or antiparallel to the momentum. The weak chiral projectors are nevertheless well defined for the field.

[a]Lecture timestamp: 00:42:44.

A massive particle can be slowed to rest and accelerated in the opposite direction without reversing its spin. Its helicity then changes sign. A massless particle cannot be overtaken or brought to rest by a Lorentz boost, so its helicity is invariant under proper boosts. This motivates the lecture's statement that a species with only one physical helicity must be massless. The more exact field-theory statement is that a Lorentz-invariant Dirac mass requires both chiralities.

Question from the audience. A circularly polarized photon can reflect from a mirror and reverse direction without ever coming to rest. Does that contradict the helicity argument?[a]

Response. No. The mirror is an external system that can exchange angular momentum and reverse the propagation direction. A reflection process is not a Lorentz boost that overtakes the photon. Whether a channel exists also depends on the allowed polarization states and on the material boundary conditions.

[a]Lecture timestamp: 00:45:28.

Question from the audience. Does an ordinary electron beam contain one handedness or both?[a]

Response. Both in general. A massive electron state contains both chiral components, and an unpolarized beam contains both helicities. Specialized sources can produce a polarized beam with a large helicity asymmetry.

[a]Lecture timestamp: 00:46:10.

Question from the audience. Can a beam be focused independently of its spin, or do magnetic fields separate the two polarizations?[a]

Response. Ordinary position focusing can be largely insensitive to spin, but magnetic moments couple to magnetic-field gradients. Spin-selective sources, Stern–Gerlach-like gradients, and accelerator spin dynamics can therefore prepare or separate polarizations.

[a]Lecture timestamp: 00:46:29.

Question from the audience. Is the weak difference between left and right merely an accident?[a]

Response. For the present course it is an experimental fact built into the Standard Model. It is too systematic to feel like a casual accident, but the Standard Model itself does not derive the complete pattern of chiral charges from a deeper principle.

[a]Lecture timestamp: 00:48:13.

1.4 The Yukawa Construction

1.4.1 Let the Higgs carry the missing charge

The symmetry forbids a numerical mass, but it allows an interaction with a field that supplies the missing transformation phase. Introduce a complex scalar with

$$\phi \longmapsto e^{i\theta}\phi, \qquad \phi^\dagger \longmapsto e^{-i\theta}\phi^\dagger. \tag{1.20}$$

Then

$$\mathcal{L}_Y = -y\,\bar{\psi}_L\phi\psi_R - y^*\,\bar{\psi}_R\phi^\dagger\psi_L \tag{1.21}$$

is gauge invariant. In the first term, $\bar{\psi}_L$ contributes $e^{-i\theta}$ and ϕ contributes $e^{i\theta}$; the phases cancel. The second term is its Hermitian conjugate.

Equivalently, the chiral equations acquire field-dependent mixing:

$$i\bar{\sigma}^\mu D_\mu\psi_L = y\phi\psi_R, \tag{1.22}$$

$$i\sigma^\mu\partial_\mu\psi_R = y^*\phi^\dagger\psi_L. \tag{1.23}$$

Before symmetry breaking, this is an interaction rather than a mass. A right-chiral fermion can become a left-chiral fermion only while emitting or absorbing the scalar quantum required to balance the charge. The coefficient y is a Yukawa coupling, not the gauge charge g.

Question from the audience. Is it legitimate simply to put ϕ and $\phi^\dagger$ on the right-hand sides of the two equations? [a]

Response. It is legitimate precisely because their transformation phases repair the mismatch. The decisive test is not the visual form of the equation but gauge invariance of the interaction $\bar{\psi}_L\phi\psi_R + \text{h.c.}$.

[a]Lecture timestamp: 00:51:55.

1.4.2 The vacuum converts interaction into mass

Now identify ϕ with the Higgs field and let it settle at nonzero vacuum magnitude. In the lecture's simplified normalization,

$$\phi \simeq f, \tag{1.24}$$

so (1.21) contains

$$\mathcal{L}_Y \supset -yf\,\bar{\psi}_L\psi_R - y^*f\,\bar{\psi}_R\psi_L. \tag{1.25}$$

After a harmless phase redefinition when appropriate, the fermion mass is

$$m_f = yf. \tag{1.26}$$

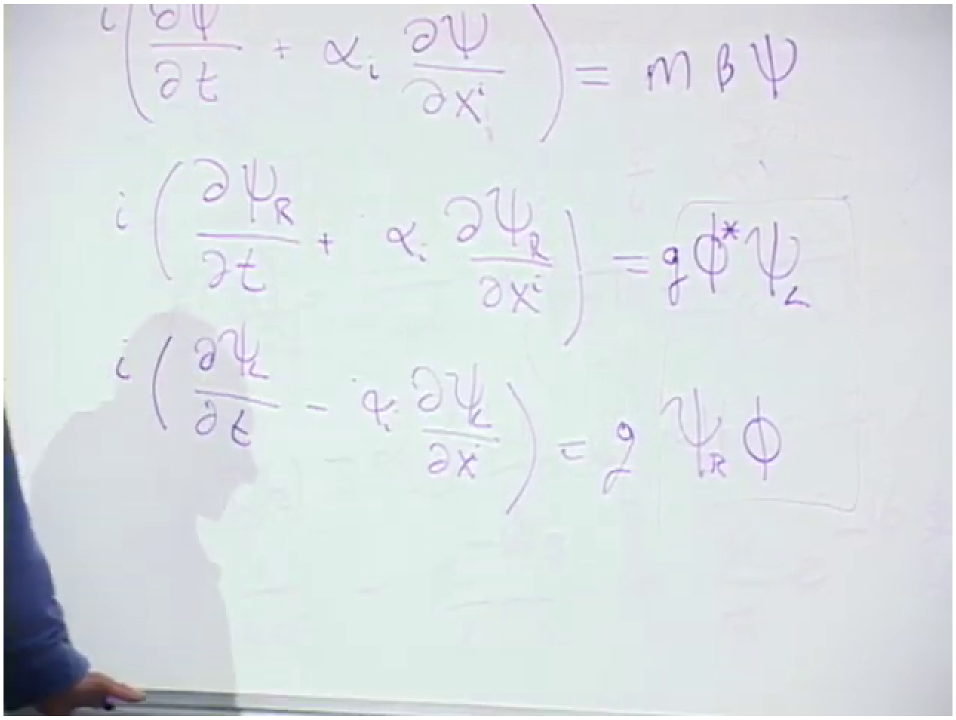

Figure 1.5: The bare mass on the right-hand sides has been replaced by the charged scalar field. The exact board signs are schematic; the invariant structure is given in (1.21).

Lecture frame: 00:51:30.

This is the fermionic counterpart of the vector result $m_A = gf$. The same Higgs vacuum that supplies a longitudinal polarization to the weak gauge bosons supplies the missing weak quantum numbers needed to join left and right fermions.

In the canonically normalized Standard Model convention,

$$H(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad v \simeq 246 \text{ GeV}, \tag{1.27}$$

and therefore

$$m_f = \frac{y_f v}{\sqrt{2}}. \tag{1.28}$$

The lecture's f and the conventional $v/\sqrt{2}$ differ only by this choice of normalization.

Question from the audience. If only the product yf is a fermion mass, how can the coupling y and vacuum scale f be determined separately? [a]

Response. The weak gauge-boson masses determine the vacuum scale once the gauge couplings are known. A fermion's measured mass then determines its Yukawa coupling. In the full theory, $m_W = gv/2$, $m_Z = v\sqrt{g^2 + g'^2}/2$, and $y_f = \sqrt{2}m_f/v$.

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[a] Lecture timestamp: 00:55:36.

Question from the audience. Is the weak-boson mass about 900 GeV? [a]

Response. No; the lecture immediately corrects the slip to roughly 90 GeV. More precisely, $m_W \simeq 80$ GeV and $m_Z \simeq 91$ GeV, while the underlying vacuum scale is $v \simeq 246$ GeV.

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[a] Lecture timestamp: 00:57:43.

1.4.3 One vacuum scale, many Yukawa couplings

Every charged fermion species has its own Yukawa coupling:

$$m_i = y_i f \qquad \text{(lecture normalization).} \tag{1.29}$$

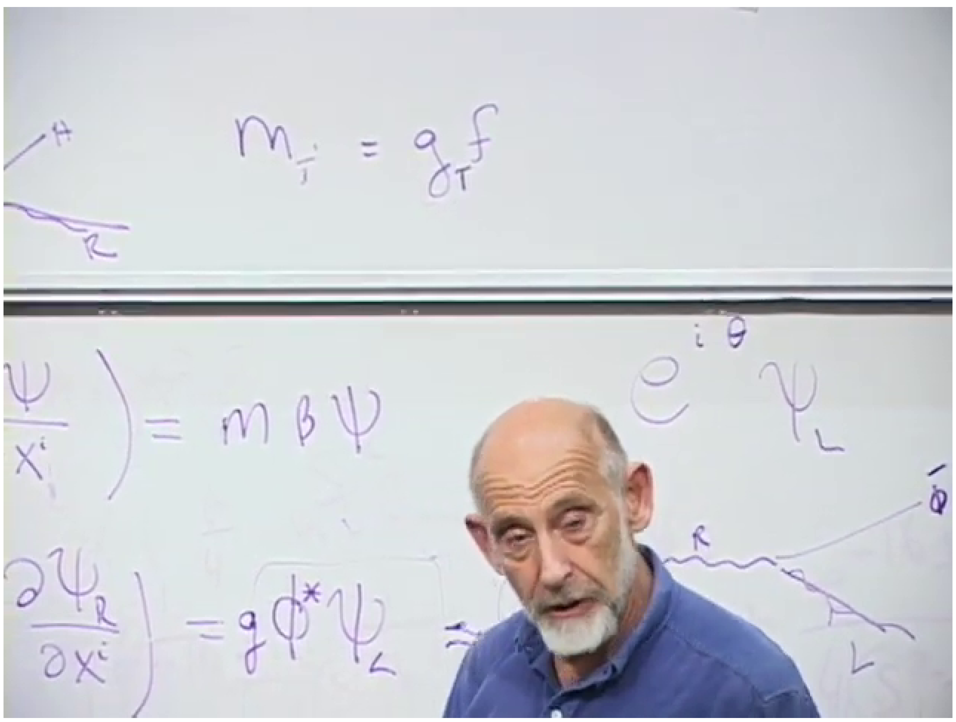

Figure 1.6: The board summarizes the species dependence as $m_i = g_i f$; the g_i written there are the Yukawa couplings called y_i here.

Lecture frame: 01:16:30.

For the electron, the lecture estimates

$$y_e \sim \frac{0.5\ \text{MeV}}{10^2\ \text{GeV}} \sim 10^{-5}. \tag{1.30}$$

Using the canonical value $v = 246$ GeV gives the more precise $y_e \simeq 2.9 \times 10^{-6}$. The top quark lies at the opposite end:

$$y_t = \frac{\sqrt{2} m_t}{v} \simeq 1. \tag{1.31}$$

Thus the Higgs mechanism explains how a gauge-invariant mass can exist, but it does not explain the enormous hierarchy among the y_i. The observed fermion masses merely translate that hierarchy into measured numbers.

Question from the audience. Does the top quark really have a Yukawa coupling of order one?[a]

Response. Yes. Its mass is comparable to the electroweak scale, so its coupling is not a tiny perturbative number like the electron's. In the modern normalization y_t is close to unity.

[a]Lecture timestamp: 00:59:09.

1.4.4 Restoring the physical Higgs fluctuation

Freezing the scalar at its vacuum value hid the radial excitation. Restore it schematically:

$$\phi(x) \simeq f + h(x). \tag{1.32}$$

The Yukawa factor becomes

$$y\phi \simeq yf + yh = m_f + yh. \tag{1.33}$$

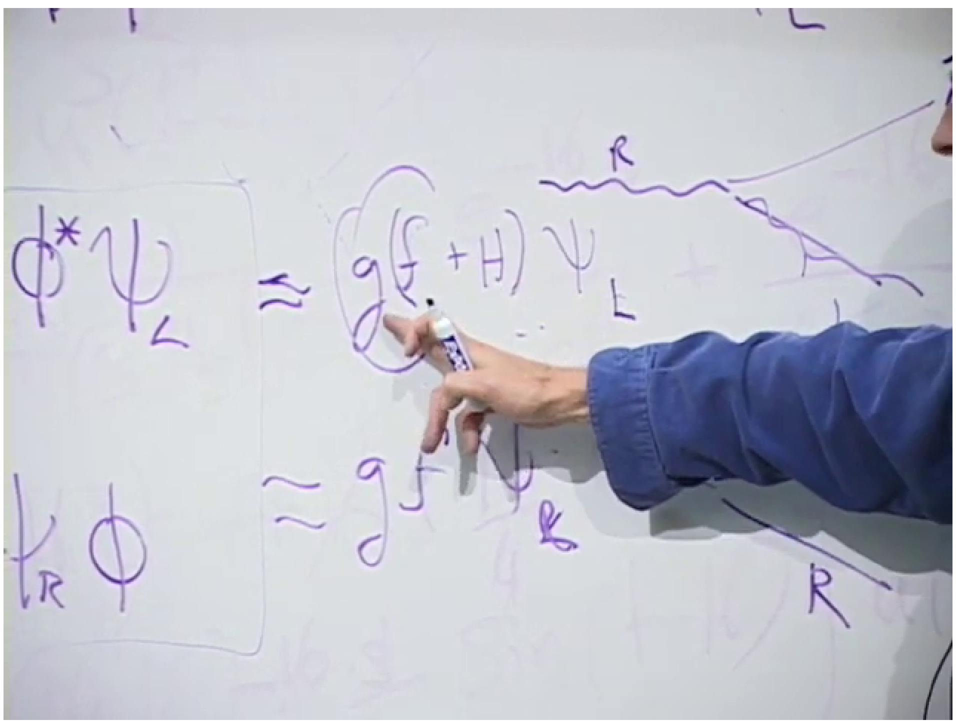

Figure 1.7: The board restores the Higgs fluctuation inside the Yukawa term: the constant part produces the fermion mass and the fluctuating part produces Higgs emission or absorption.

Lecture frame: 01:01:15.

The first term in (1.33) is the mass. The second is a vertex in which a Higgs boson couples to the fermion. In canonical Standard Model notation,

$$\mathcal{L}_Y \supset -m_f\bar{\psi}\psi - \frac{m_f}{v}h\bar{\psi}\psi. \tag{1.34}$$

The coupling is proportional to the fermion mass. Subject to kinematics, color factors, phase space, and competing channels, the Higgs therefore couples more strongly to heavier fermions.

Question from the audience. Are coupling constants themselves probabilities? [a]

Response. They enter quantum amplitudes. A simple rate is often proportional to the modulus square of an amplitude and hence to a coupling squared, but phase space, spin sums, interference, loops, and multiple vertices also matter. A coupling should not be identified directly with a probability.

[a]Lecture timestamp: 01:03:35.

Question from the audience. Does this argument assume that there is only one Higgs field?[a]

Response. It describes the minimal Standard Model, which has one complex Higgs doublet. Three of its four real components are absorbed by $W^{\pm}$ and Z, leaving one physical neutral scalar. Extensions can contain more doublets or other scalar multiplets. Supersymmetric models typically require two Higgs doublets and therefore contain several physical Higgs states.

[a]Lecture timestamp: 01:03:59.

That historical expectation can now be compared with experiment. [2]

1.5 Antiparticles, Beams, and the Meaning of Chirality

Before turning to neutrinos, the class returns to several operational questions about antiparticles and accelerator beams.

Question from the audience. What handedness does the antiparticle of a left-handed electron have?[a]

Response. A left-chiral electron field annihilates left-helicity electrons and creates right-helicity positrons in the ultrarelativistic limit. This is one place where particle versus antiparticle and chirality versus helicity must be stated carefully.

[a]Lecture timestamp: 01:06:05.

Question from the audience. Can an electron and positron collide to make a Z boson?[a]

Response. Yes, when their center-of-mass energy matches the allowed channel. Electron–positron colliders studied the Z resonance with great precision. Charge and the other quantum numbers are conserved by the full process.

[a]Lecture timestamp: 01:07:13.

Question from the audience. Are positrons all produced through one special weak process?[a]

Response. No. Pair production is a standard electromagnetic route. For example, an energetic electron can radiate a virtual photon, and that photon can produce an e^-e^+ pair in the field of another particle. The lecture sketches this with a Feynman diagram.

[a]Lecture timestamp: 01:07:37.

[2]Editorial clarification: This lecture was recorded before Higgs discovery. ATLAS and CMS announced a Higgs-like particle in 2012, now measured near 125 GeV. Its observed couplings are broadly consistent with the mass-proportional Standard Model pattern. The statement that it decays preferentially to the heaviest particles always carries the kinematic caveat: an on-shell $t\bar{t}$ pair is too heavy for a 125 GeV Higgs.

Question from the audience. Can a linear accelerator select the speed or energy of its electrons?[a]

Response. Yes. The accelerating fields and timing select an energy range. Once electrons are ultrarelativistic their speed is already extremely close to c, so additional accelerator energy mostly increases momentum and total energy rather than visibly increasing speed.

[a] Lecture timestamp: 01:09:24.

Question from the audience. Can an accelerator prepare a beam of one polarization? [a]

Response. Yes, although it is technically demanding. One can prepare spin-polarized electrons at low energy with a spin-dependent source and then accelerate them while controlling spin precession. Modern colliders quantify the resulting beam polarization rather than treating it as perfectly pure.

[a] Lecture timestamp: 01:09:47.

1.6 Neutrino Mass and the Majorana Possibility

1.6.1 Why neutrino mass forced an extension

The original minimal Standard Model included only left-chiral neutrino fields and therefore predicted massless neutrinos. Neutrino oscillations show that at least two neutrino mass eigenvalues differ and that neutrino masses are nonzero, although tiny compared with charged-fermion masses.

Question from the audience. How small are the neutrino masses, and where is their right-handed component?[a]

Response. Oscillations determine mass-squared differences, not the absolute masses. The relevant splittings correspond to scales of roughly 0.009 eV and 0.05 eV. A Dirac description adds sterile right-chiral neutrinos; a Majorana description instead relates the missing chirality to charge-conjugate neutrino fields.

[a] Lecture timestamp: 01:10:38.

For a Dirac neutrino one adds a new field ν_R and repeats the Yukawa construction:

$$\mathcal{L}_{\nu,\mathrm{Dirac}} = -y_\nu \bar{L}\, \widetilde{H}\, \nu_R + \text{h.c.}, \qquad m_\nu = \frac{y_\nu v}{\sqrt{2}}. \tag{1.35}$$

The required y_ν is extraordinarily small.

There is another logical possibility because the neutrino carries no electric charge. A left-chiral neutrino can be coupled to the charge conjugate of itself, which has the opposite chirality:

$$\nu_L \longleftrightarrow (\nu_L)^c_R. \tag{1.36}$$

A mass eigenstate built this way is a Majorana fermion: it is its own antiparticle, up to phase conventions. The analogous electron–positron mixing is forbidden because it would violate exact electric-charge conservation.

Question from the audience. Why is particle–antiparticle mixing allowed for a neutrino but not for an electron?[a]

Response. An electron and a positron have opposite electric charge, so a mass term that mixes them as the same one-particle state would violate the unbroken electromagnetic symmetry. A neutrino is electrically neutral. If no exact conserved lepton number distinguishes it from its antiparticle, a Majorana mass is allowed.

[a]Lecture timestamp: 01:13:43.

The name honors Ettore Majorana, the young Italian physicist who formulated this kind of real fermion representation and disappeared in 1938. The lecture pauses over the historical mystery: his fate was never established. [3]

1.7 Why the Familiar Spectrum Clusters Near the Electroweak Scale

1.7.1 The common scale and its important qualifications

Question from the audience. Why was the Higgs mass expected to be reachable at the weak scale?[a]

Response. The curvature of the Higgs potential sets the radial Higgs mass, while the same vacuum scale f sets the W and Z masses. Without an extreme scalar self-coupling, one therefore expected the Higgs mass to be of the same broad order as the weak scale. The observed 125 GeV value indeed lies there.

[a]Lecture timestamp: 01:14:48.

The lecture's striking summary is that the elementary masses of the minimal Standard Model trace back to one broken vacuum:

$$m_W = \frac{1}{2} g v, \tag{1.37}$$

$$m_Z = \frac{1}{2}\sqrt{g^2 + g'^2}\, v, \tag{1.38}$$

$$m_f = \frac{1}{\sqrt{2}} y_f v. \tag{1.39}$$

If v vanished while the dimensionless couplings were held fixed, the weak gauge bosons and charged elementary fermions would be massless.

That statement must not be enlarged beyond its domain. The photon and gluons remain massless because an unbroken gauge symmetry protects them. Most of the proton and neutron mass is QCD binding energy, not the sum of their quark Higgs masses. Neutrino mass requires an extension of the minimal model. The Higgs mass itself depends on the scalar potential and is not simply another Yukawa mass.

[3]Editorial clarification: A gauge-invariant Majorana mass for the active Standard Model neutrino is not a renormalizable bare term before electroweak breaking. At low energy it can arise from the dimension-five Weinberg operator $(LH)(LH)/\Lambda$, or through a seesaw involving heavy sterile neutrinos. Whether neutrinos are Dirac or Majorana particles remains experimentally unsettled.

1.7.2 Particles whose mass need not wait for symmetry breaking

Suppose a new fermion had left- and right-chiral components with identical gauge quantum numbers. Then $\bar{\Psi}_L \Psi_R + \text{h.c.}$ would already be gauge invariant. Such a *vectorlike* fermion could possess a bare mass M unrelated to v:

$$\mathcal{L}_{\text{vectorlike}} = -M\bar{\Psi}\Psi. \tag{1.40}$$

There is no electroweak symmetry obstruction to $M \gg v$.

This motivates the lecture's broad naturalness argument. Particles protected by gauge or chiral symmetry remain massless until the Higgs vacuum breaks that protection, so their masses are tied to the relatively small scale v. Particles with no such protection may naturally be much heavier and therefore absent from the observed low-energy spectrum. What we see may be a selection effect: the accessible particles are the ones whose masses were forced to wait for electroweak symmetry breaking.

Question from the audience. Would particles with independent, much larger masses play no role in reality?[a]

Response. They would play little role in ordinary low-energy laboratory physics because they are hard to produce, but they could still shape high-energy history or cosmology. The lecture offers dark matter as a possible place where a heavier hidden spectrum might first reveal itself.

[a] Lecture timestamp: 01:19:39.

The 2010 discussion reflects the then-popular expectation of weak-scale or somewhat heavier dark matter. That remains a useful model class, not an established fact. Dark matter has been detected gravitationally, but its particle identity and the origin of its mass remain unknown.

1.8 Closing Questions: Cosmic Rays, Inertia, Gravity, and Chirality

1.8.1 Mass is not high laboratory energy

Question from the audience. What is the mass of a cosmic ray? [a]

Response. A cosmic ray may be a proton, an atomic nucleus, an electron, or a photon; its invariant mass is the mass of that particle or system. What is spectacular is its laboratory energy, which for the highest events reaches roughly 10^{20} to 10^{21} eV. High energy is not the same as high invariant mass.

[a] Lecture timestamp: 01:20:57.

A cosmic ray strikes a target particle that is approximately at rest. For a projectile proton of laboratory energy $E \gg m_p$,

$$s = (p_1 + p_2)^2 \simeq 2m_pE + 2m_p^2, \qquad \sqrt{s} \simeq \sqrt{2m_pE}. \tag{1.41}$$

Only the square root of the projectile energy becomes center-of-mass energy. For equal head-on beams of energy E_{beam},

$$\sqrt{s} \simeq 2E_{\text{beam}}. \tag{1.42}$$

This is why a collider is far more efficient than a fixed target at converting controlled beam energy into new-particle mass.

Question from the audience. Why can CERN's collider be a more potent probe than a single much more energetic particle hitting matter at rest? [a]

Response. In a fixed-target collision, momentum conservation carries most of the laboratory energy forward in the motion of the final state. In a symmetric collider the total spatial momentum can vanish, leaving the beam energy available as center-of-mass energy. The very highest cosmic rays can still exceed the LHC in $\sqrt{s}$, but they arrive unpredictably and cannot provide collider luminosity or a controlled detector environment.

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[a] Lecture timestamp: 01:21:50.

1.8.2 Rest mass and resistance to acceleration

Question from the audience. Are inertia and mass essentially the same thing? [a]

Response. At rest, invariant mass is both rest energy through $E_0 = mc^2$ and the parameter governing the low-velocity response to a force. Special relativity explains why those definitions belong to one quantity. For moving bodies it is clearer to keep invariant mass fixed and describe acceleration using relativistic energy and momentum.

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[a] Lecture timestamp: 01:22:41.

The invariant relation

$$E^2 = p^2c^2 + m^2c^4 \tag{1.43}$$

contains both meanings. Near rest,

$$E = mc^2 + \frac{p^2}{2m} + \cdots , \tag{1.44}$$

so the same m that gives the rest energy appears in the kinetic response. At high speed the relation between force and acceleration depends on whether the force is parallel or perpendicular to the velocity. This directional dependence is why modern treatments avoid a velocity-dependent *relativistic mass* and reserve mass for the Lorentz invariant m.

1.8.3 What gravity couples to

Question from the audience. Once the Higgs gives a particle mass, does it then begin to couple to gravity?[a]

Response. Its rest energy contributes to gravity, but mass is not the whole source. Photons are massless and are nevertheless deflected by gravity. In general relativity the source is the stress–energy tensor: energy density, momentum flow, pressure, and stress. For slowly moving matter the dominant contribution is its rest-mass energy.

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[a] Lecture timestamp: 01:23:58.

The bending of light near the Sun makes the point operationally. A photon responds to spacetime curvature despite having $m = 0$, and its energy and momentum also contribute to the gravitational field. For two nearly static massive bodies, energy is dominated by mc^2, so Newtonian gravity can be written in terms of their masses without displaying the more general stress–energy statement.

Question from the audience. Is a high-energy particle harder to accelerate in the same way in every direction?[a]

Response. No. The change of momentum for a force parallel to the motion differs from the change for a transverse force. It is correct that a more energetic particle is harder to deflect, but assigning it one enlarged moving mass hides this directional structure.

[a]Lecture timestamp: 01:25:20.

1.8.4 An electron at rest and the promised distinction

Question from the audience. How can an electron at rest be called left- or right-handed?[a]

Response. It cannot be assigned helicity at rest because helicity uses the direction of momentum. A rest spinor can still be decomposed into left- and right-chiral components. For a massive electron both components are present, and the mass term couples them.

[a]Lecture timestamp: 01:26:32.

Question from the audience. If an electron at rest has no helicity, can it still participate in a weak interaction?[a]

Response. Yes. The weak interaction couples to chirality, not to a frame-dependent helicity label. At high energy helicity is a convenient approximation to chirality; near rest that shortcut fails. This is the subtle distinction the lecture had postponed.

[a]Lecture timestamp: 01:27:18.

1.9 Historical End-of-Lecture Questions

1.9.1 The pre-discovery Higgs window

In 2010 the Higgs had not yet been seen. Precision electroweak observables, especially loop-sensitive relations among the W, Z, top, and other parameters, favored a comparatively light Higgs within the Standard Model. The lecture quotes a likely upper region near 160 GeV, well below v but not separated from it by an order of magnitude.

Question from the audience. What evidence then suggested that the Higgs was lighter than the 250 GeV vacuum scale?[a]

Response. High-precision measurements could be compared with radiative corrections whose values depend logarithmically on the Higgs mass. The resulting global fits favored a light Higgs. This was indirect evidence, not a direct mass measurement; the later direct value is about 125 GeV.

[a]Lecture timestamp: 01:28:28.

Question from the audience. Could several minima of the Higgs potential produce different values of f, perhaps tied to different particle generations? [a]

Response. Different minima can support very different low-energy physics, but there is no simple inference from the observed generations to multiple Higgs minima. The lecture defers the broader subject of multiple vacua to a later course rather than pretending to settle it here.

[a] Lecture timestamp: 01:29:28.

Question from the audience. Was the Higgs expected to be the first new particle found at the LHC?[a]

Response. Not necessarily. Its visibility depends on production rates and decay channels, some of which can be difficult or invisible. Before LHC data, supersymmetric or other new particles could plausibly have appeared first. Historically, the Higgs was discovered and no supersymmetric particle has yet been established.

[a] Lecture timestamp: 01:31:32.

1.9.2 When a neutral particle is its own antiparticle

Question from the audience. Do the photon and gluon lack antiparticles because they are massless?[a]

Response. No. Masslessness is not the criterion. A photon is its own antiparticle because no conserved charge distinguishes a photon from an antiphoton. Each gluon is also a self-conjugate gauge quantum in the appropriate color representation, although gluons themselves carry color and the color labels mix under charge conjugation. A neutron is electrically neutral but has a distinct antineutron because baryon number and its quark content distinguish them.

[a] Lecture timestamp: 01:32:12.

The neutrino returns as the subtle case. It has weak interactions, and the ultrarelativistic states called neutrino and antineutrino interact with opposite helicities. If an exact lepton number distinguishes them, they can form a Dirac particle with a distinct antiparticle. If lepton number is not exact and the mass is Majorana, the massive particle is its own antiparticle even though its two helicity states behave differently in weak processes.

Question from the audience. Do weak charge and lepton number prove that neutrinos and antineutrinos must be distinct?[a]

Response. Not conclusively. Weak interactions distinguish the helicity states, but a Majorana mass can combine a neutrino field with its charge conjugate. The decisive issue is whether an exact conserved quantum number forbids that mixing. Total lepton number is not imposed as an exact gauge symmetry by the Standard Model.

[a] Lecture timestamp: 01:32:46.

Ordinary flavor oscillations require a separate caution. [4]

[4] Editorial clarification: Ordinary observed flavor oscillations mix ν_e, ν_μ, ν_τ of the same neutrino or antineutrino sector; they do not by themselves demonstrate neutrino–antineutrino conversion. Lepton-number-violating processes such as neutrinoless double-beta decay would provide direct evidence for Majorana mass, but none has yet been confirmed.

Question from the audience. Does the Standard Model explain why electric charges are quantized?[a]

Response. The lecture acknowledges the question but does not develop it. Within the Standard Model, gauge invariance and anomaly cancellation strongly constrain consistent hypercharge assignments; a deeper explanation often invokes grand unification or other ultraviolet structure.

[a]Lecture timestamp: 01:33:39.

1.10 What the Lecture Has Established

The gauge-boson and fermion stories now line up.

1. A derivative-only gauge theory has no energy gap for a homogeneous vector potential. Gauge symmetry forbids a bare vector mass.
2. A charged scalar with nonzero vacuum magnitude contributes $g^2 f^2 A_\mu A^\mu$. Its angular Goldstone mode becomes the longitudinal polarization of a massive weak gauge boson.
3. A Dirac mass couples ψ_L and ψ_R. Because the weak interaction assigns them different gauge quantum numbers, a bare left–right mass term is forbidden.
4. A Yukawa interaction inserts the Higgs field between the two chiralities and is gauge invariant. Replacing the Higgs by its vacuum value gives $m_f = y_f f$, while retaining its fluctuation gives the physical Higgs–fermion coupling.
5. Neutrino mass obeys the same left–right logic but admits both Dirac and Majorana realizations because the neutrino is electrically neutral.
6. Particles without gauge or chiral protection could possess masses unrelated to the electroweak scale and may lie far above present experiments.

The unifying statement is therefore not that the Higgs is a substance that mechanically drags on particles. It is that the electroweak vacuum has a nonzero scalar order parameter. That order parameter changes which gauge-invariant terms are possible in the low-energy equations. For vectors it opens a zero-momentum gap; for fermions it permits left–right mixing. The different masses then measure different couplings to the same broken vacuum.